Advanced Requirements Engineering: 2025-10-30 questions and suggested solutions

Part 1: proposition/reason questions

Each question is worth 1 point. Select one alternative:

Answer	Meaning
A	Proposition and reason are true, and the reason explains the proposition.
B	Both are true, but the reason does not explain the proposition.
C	Proposition true; reason false.
D	Proposition false; reason true.
E	Both false.

A warning beside an answer takes precedence over any apparent certainty of its letter.

1.1 (1 point; PDF p. 2)

Proposition: According to IEEE Definition of Requirement from 1990, a requirement must be explicitly written down, otherwise it is not a requirement.
Reason: According to the IEEE Definition from 1990, a Requirement is: (1) A condition or capability needed by a user to solve a problem or achieve an objective. (2) A condition or capability that must be met or possessed by a system or system component to satisfy a contract, standard, specification, or other formally imposed document. (3) A documented representation of a condition or capability as in (1) or (2).
Answer: D. The quoted definition includes needs and capabilities as well as their documented representations. Writing is not a necessary condition for something to be a requirement.

1.2 (1 point; PDF p. 3)

Proposition: Tacit knowledge is difficult to transfer to another person by writing it down or verbalizing it.
Reason: Developers and users may not share a common language, thus making it hard to communicate about requirements.
Answer: B. Both statements are true. Tacit knowledge is hard to articulate even when people share a language; the reason describes a gulf of understanding rather than explaining tacitness.

1.3 (1 point; PDF p. 3)

Proposition: Only real persons can be stakeholders.
Reason: A stakeholder must have a valid id with their contact information.
Answer: E. Stakeholders may include groups and organisations. Possession of an identity document is not a condition for being a stakeholder.

1.4 (1 point; PDF p. 4)

Proposition: According to Lauesen, interviews are not suitable for conflict resolution.
Reason: Interviews must always be conducted with a single individuum as interviewee.
Answer: C. In the course comparison, interviews help discover conflicts but are not the technique for negotiating their resolution. Interviews can involve groups, so the reason is false.

1.5 (1 point; PDF p. 4)

Proposition: According to Lauesen, each group that participates in a focus group must "get something".
Reason: If a stakeholder group does not get something in return, they are rarely willing to contribute to the system.
Answer: A. The intended rationale is that each participating group needs to see a benefit to remain willing to contribute.

> **Ambiguity in Solution:** An older version of this question allowed to argue that the reason does not fully support the proposition. Even with this question, we encourage to discuss this in an exam review.

1.6 (1 point; PDF p. 5)

Proposition: Goal domain tracing is both useful for elicitation and validation of requirements.
Reason: Goal domain tracing allows to validate whether each task or quality factor has a purpose with respect to business goals as well as to validate that each business goal is sufficiently supported by tasks and quality factors.
Answer: B. Both statements are true. The reason discusses validation; it does not explain the elicitation part of the proposition.

1.7 (1 point; PDF p. 6)

Figure supplied in the PDF: the WRSP world/interface/machine diagram places assumptions W and requirements R in the world, specification S at the shared interface, and program P and machine M on the machine side; it distinguishes hidden and visible environment/system phenomena (`e_h`, `e_v`, `s_v`, `s_h`). See the linked PDF for the original graphic.
Proposition: The WRSP model does support architectural decomposition.
Reason: The WRSP model defines requirements and specification based on events in the "world" and states in the "machine". These states can be used to describe the logical architecture, thus allowing to map requirements to logical components.
Answer: D. WRSP distinguishes the world, interface and machine but does not directly support architectural decomposition. The reason describes how its states and events can inform a logical architecture and the mapping of requirements to components. This requires additional modelling decisions; it is not decomposition support provided by WRSP itself.

1.8 (1 point; PDF p. 7)

Proposition: According to Glinz, an attribute is a performance requirement, a specific quality requirement, or a constraint.
Reason: It is equally difficult to make performance requirements, specific quality requirements, and constraints measurable.
Answer: E. In the course taxonomy, an attribute is a performance or specific quality requirement; a constraint is a separate category. These categories also differ in how readily they can be made measurable.

1.9 (1 point; PDF p. 7)

Proposition: Lauesen discourages using requirements as part of a contract between customer and supplier.
Reason: If requirements are part of a contract between customer and supplier, changing a requirement entails changing the contract, which may involve legal teams on both sides.
Answer: D. Requirements may form part of a customer/supplier contract. Contractual change can require renegotiation and legal involvement; that does not imply that Lauesen discourages contractual requirements.

1.10 (1 point; PDF p. 8)

Proposition: It is hard to ensure that a requirements specification is complete.
Reason: Completeness of a requirements specification refers to all necessary requirements being included.
Answer: B. Both statements are true. Defining completeness does not by itself explain why establishing it is difficult: missing needs, exceptions and tacit knowledge are hard to discover.

1.11 (1 point; PDF p. 8)

Proposition: It is insufficient to define completeness of a specification in relation to the template for the specification.
Reason: Lauesen warns about template blindness, which can narrow down ones view to what the template deals with.
Answer: A. Filling every template section does not establish that every relevant need is covered. Template blindness explains why the template can conceal omissions.

1.12 (1 point; PDF p. 9)

Proposition: A feature based approach to documenting requirements facilitates an unambiguous requirements specification.
Reason: In a feature based approach, each requirements is one statement about a feature of a system.
Answer: D. Individual feature statements can still have multiple interpretations. The reason describes the feature style, which does not guarantee unambiguous communication.

1.13 (1 point; PDF p. 9)

Proposition: In order to achieve a good enough requirements specification, it is important to focus on the requirements that everybody knows about.
Reason: Requirements that are well known by many stakeholders are easily misinterpreted.
Answer: E. The intended answer prioritises requirements most at risk of being misunderstood or overlooked. Widely shared understanding alone is not a reason to prioritise detailed specification.

> **Changed wording / recorded ambiguity:** newer version of this question says "one should primarily focus", whereas this version of the exam says "it is important to focus". Without "primarily", the proposition is less clearly false; C is arguable if it merely says shared requirements also deserve attention. In the 2025 exam, we offered 0.5P for answering C in cases where this made a difference on the grade.

1.14 (1 point; PDF p. 10)

Proposition: Data expressions are suitable for specifying message protocols.
Reason: Data expressions are simplified screens with realistic data.
Answer: C. Data expressions can specify structured messages and protocols. Simplified screens containing realistic data describe virtual windows, not data expressions.

1.15 (1 point; PDF p. 10)

Proposition: The event list & function list approach allows to trace between product-level requirements and design-level requirements.
Reason: The event list & function list approach establishes many-to-many relationships between domain events and product events.
Answer: D. The approach relates domain events to product functions/events, rather than product-level requirements to design-level requirements. Several domain events may use several product functions.

1.16 (1 point; PDF p. 11)

Proposition: To a degree, scenario-based approaches such as task descriptions can support completeness of functional requirements.
Reason: It is relatively easy to spot missing sub- tasks in task descriptions
Answer: A. Walking through user tasks makes omitted steps and variants easier to notice, helping assess coverage of functional requirements.

1.17 (1 point; PDF p. 11)

Proposition: Creativity techniques in requirements engineering should replace elicitation.
Reason: Stakeholders cannot provide novel requirements.
Answer: E. Creativity complements elicitation. Stakeholders can contribute novel requirements and ideas; they are not restricted to reporting existing needs.

1.18 (1 point; PDF p. 12)

Proposition: Convergent thinking (to evaluate and choose the best ideas) should be avoided during Creative Problem Solving (CPS).
Reason: A core principle of CPS is to defer judgment.
Answer: D. CPS uses divergent generation and convergent evaluation. Deferring judgment applies while generating ideas; evaluation is needed later.

1.19 (1 point; PDF p. 12)

Proposition: A taxonomy of system and software quality characteristics helps to identify relevant quality attributes.
Reason: Many established quality taxonomies such as ISO25010 and ISO25019 are incomplete.
Answer: B. In the course interpretation, quality taxonomies are useful checklists but do not exhaust every context-specific concern. Their incompleteness does not explain their usefulness.

1.20 (1 point; PDF p. 13)

Proposition: Functionality is not a quality characteristic.
Reason: Quality characteristics include specific quality attributes and performance, but no functional requirements.
Answer: D. The notes distinguish quality characteristics from functional requirements: functionality can be a quality characteristic, as shown in the lecture taxonomy, even though individual functional requirements are not quality requirements.

1.21 (1 point; PDF p. 13)

Proposition: Open metric allows to rely on the suppliers expertise with respect to knowing how to measure the quality of the system.
Reason: Open metric delegates responsibility for reaching the project goals to the supplier, who is now responsible for the acceptance test of quality requirements.
Answer: C. Open Metric lets the supplier propose how quality is measured or reported. It does not transfer responsibility for the customer's business goals or acceptance decision to the supplier.

1.22 (1 point; PDF p. 14)

Proposition: Planguage is a template that facilitates testable specification of quality requirements.
Reason: While the other fields in the Planguage template provide valuable meta data, it is the Meter field that ensures testability of requirements.
Answer: C. Meter alone is insufficient. A measurement procedure also needs a defined scale and acceptance target to support a pass/fail decision.

> **Recorded ambiguity:** One could argue that Meter helps. Meter alone cannot supply a threshold, but the wording does not explicitly say "alone". We have given partial points for answering 'A' in specific cases where it made a difference.

1.23 (1 point; PDF p. 14)

Proposition: Priority grouping is a simple, but often flawed approach to prioritization.
Reason: Priority grouping does not require challenging requirements against each other and may lead to too many high priority requirements.
Answer: A. Independent priority labels do not force comparison or trade-offs, so many requirements can end up in the highest group.

1.24 (1 point; PDF p. 15)

Proposition: In prioritization, the ordinal scale should be favored, since it has the highest expressive power of all available scales.
Reason: The ordinal scale allows to express that requirement A is 2x as important than requirement B.
Answer: E. An ordinal scale provides order, not meaningful ratios. Saying that one requirement is twice as important requires a ratio scale.

1.25 (1 point; PDF p. 15)

Proposition: Horizontal traceability refers to tracing across different phases of the waterfall model.
Reason: Ramesh and Edwards define horizontal traceability as tracing between artifacts of different types (e.g. tests and requirements).
Answer: E. Use the terminology in this course: horizontal links connect artifacts of the same type; vertical links connect different types. The quoted attribution in the reason is therefore false in the course terminology.

1.26 (1 point; PDF p. 16)

Proposition: Pre-requirements traceability supports validation.
Reason: Pre-requirements traceability connects requirements to their source or origin, thus establishing whether a requirement is needed.
Answer: A. Linking a requirement to its originating need or goal helps validate that it is justified. Validation also includes other checks, but traceability supports it.

> **Recorded review qualification:** the examiner preferred A but recorded half a point for B in a reviewed case, acknowledging that validation also involves completeness.

1.27 (1 point; PDF p. 16)

Proposition: In change impact analysis, traceability can be used to generate the Candidate Impact Set.
Reason: If important tracelinks are missing, the candidate impact set my not include all artifacts of the actual impact set.
Answer: B. Both statements are true. Missing links explain a limitation of the candidate set, rather than why traceability can generate that set.

1.28 (1 point; PDF p. 17)

Proposition: GDT is a specific traceability matrix.
Reason: Traceability matrices are often shown with the same artifacts in both dimensions and an indication in a "cell" if the artifacts of this cell's column and row are connected.
Answer: B. GDT is a specialised traceability matrix; a general description of matrices does not explain that classification.

> **Recorded ambiguity** One might argue that we do not really know how often traceability in matrixes is done with the same artifacts in both dimensions. A clearer formulation could have been desirable. However, the lecture does point to seminal papers that imply this "classic" traceability matix as the default.

1.29 (1 point; PDF p. 17)

Proposition: A graphical model that connects business goal 1 with task description 7 is not a TIM.
Reason: A TIM is the metamodel that defines which tracelinks between artifacts are allowed and which tracelinks between artifacts are required.
Answer: A. A link between two particular artifacts is an instance. A traceability information model (TIM) defines artifact types, link types and required/allowed relationships.

1.30 (1 point; PDF p. 18)

Proposition: Even outdated tracelinks are better than no tracelinks.
Reason: Outdated tracelinks create additional work, as irrelevant artifacts must be considered when relying on traceability. At the same time, relevant artifacts may be overlooked, since a tracelink has not yet been set. In particular the latter can be critical in many use cases.
Answer: D. Outdated links can misdirect impact analysis and hide affected artifacts. These risks refute the blanket claim that outdated links are always better.

1.31 (1 point; PDF p. 18)

Proposition: A CRUD matrix supports verification of requirements.
Reason: Verification can be expressed as the question "Do the requirements fulfil the stakeholder needs/goals" and a CRUD matrix allows to trace from data requirements to goals.
Answer: C. A CRUD matrix checks consistency and coverage between data and tasks/functions. Matching stakeholder needs is validation, and CRUD does not directly map data to goals.

1.32 (1 point; PDF p. 19)

Proposition: Inspections can provide both validation and verification of requirements.
Reason: Inspections are a systematic method for manually reading through specifications.
Answer: B. Inspections can examine specification quality and, with appropriate stakeholder participation, fitness to needs. Merely describing systematic reading does not explain both uses.

1.33 (1 point; PDF p. 19)

Proposition: Usability tests is a good approach for requirements verification.
Reason: During a usability check, the facilitator will guide the user through checking the consistency of requirements.
Answer: E. The intended distinction is that usability testing validates task support with representative users. The facilitator observes rather than guiding the user through a requirements-consistency check.

1.34 (1 point; PDF p. 20)

Proposition: Agile software development implies that development teams will have to do "Just-in-time" RE.
Reason: The only other option would be to fully analyse the requirements before the team starts to develop, effectively establishing a waterfall process.
Answer: A. In the intended course argument, agile teams elaborate requirements as work approaches rather than completing all detailed analysis up front. Early breadth-first analysis still complements this practice.

1.35 (1 point; PDF p. 20)

Proposition: A user story is more than just the text, it should also include confirmation, i.e. describing its acceptance test.
Reason: The most common template for a user story (As a <role> I want <feature> so that <value>) ensures that a user story is testable.
Answer: C. Confirmation/acceptance criteria are part of the story practice. The role-feature-value sentence alone does not specify a testable acceptance condition.

1.36 (1 point; PDF p. 21)

Proposition: In large-scale agile approaches, non- functional requirements should not be managed on team level.
Reason: Non-functional requirements are usually cross-cutting, thus, they affect multiple components, functional requirements, and teams.
Answer: A. The intended answer concerns cross-team ownership and coordination of cross-cutting qualities. Teams still refine, implement and test their local contributions; "not managed on team level" should not mean teams ignore them.

1.37 (1 point; PDF p. 22)

Proposition: Many modern vehicles (not only those with self-driving capabilities) take pictures and videos of their surroundings, especially of humans that approach them while they are parked. This can be considered as an ethical problem.
Reason: Imagine a school child approaching a parked vehicle. The vehicle taking pictures and sharing them with an unknown group of recipients can be considered a violation of the EU ethical guideline of Nonmaleficience.
Answer: C. Recording and sharing images of people without their knowledge raises ethical concerns. The intended distinction is that the example primarily concerns privacy and data protection, rather than nonmaleficence (avoiding harm).

> **Possible ambiguity** The question should be defined more clearly with categories of ethical guidelins in mind. In the current formulation, full points should be granted for 'A', since these concerns can overlap: unwanted recording and sharing may also cause harm.

1.38 (1 point; PDF p. 23)

Proposition: During requirements analysis, anonymisation is no longer a concern.
Reason: Anonymisation (defined as a component of confidentiality involving not revealing any data that can be used to identify subjects) should happen during elictation.
Answer: D. Confidentiality needs attention during elicitation and throughout analysis. Handling data earlier does not end the duty to protect identities later.

1.39 (1 point; PDF p. 23)

Proposition: Ambiguous requirements can cause ethical problems.
Reason: Ambiguity in requirements often leads to ambiguous or biased interpretation, however, biases and fairness are important ethical consideration.
Answer: A. Ambiguity can enable inconsistent or biased interpretations, creating unfair outcomes. This explains the potential ethical problem.

1.40 (1 point; PDF p. 24)

Proposition: Optimizing for overall safety may still disadvantage certain groups (e.g., cyclists, children, or pedestrians in low- visibility conditions).
Reason: Without additional effort for these groups, the overall safety strategy may not sufficiently represent these groups because a) they may be underrepresented in the data (particular in low-visibility conditions) and b) they are more vulnerable than other traffic participants.
Answer: A. Aggregate safety optimisation can hide worse outcomes for underrepresented or vulnerable groups. The reason explains why group-specific evaluation and mitigation may be needed.

Part 2: interactive questions

2.1: requirements levels (10 points; PDF p. 25)

Question: Match each requirement with its most suitable level. Alternatives: Goal-level, Domain-level, Product-level, Design-level. One selection per row.

PDF row	Requirement	Solution	Why
1	The product shall support estimations that match actual scores within a standard deviation of 1%.	**Goal-level**	States a desired outcome/accuracy rather than a user task or a concrete function; follows the pool classification.
2	The system shall have a mini keyboard with start/stop button.	**Design-level**	Prescribes a particular physical interface.
3	The product shall handle the booking task.	**Domain-level**	Names a user/domain task to support.
4	The product shall have a function for storing new items based on their associated keywords.	**Product-level**	Specifies a product function and its data.
5	The product shall have a function for retrieving items based on specified keywords.	**Product-level**	Specifies a product function and its data.
6	The product shall support the user registration task.	**Domain-level**	Names a user/domain task without prescribing its interface.
7	The menu points and buttons shall work as described in appendix yy.	**Design-level**	Prescribes interface behaviour through a referenced design.
8	The system shall allow efficient usage with one hand only.	**Product-level**	Constrains product interaction without choosing a concrete control design; follows the pool classification.
9	For all fields, the user shall be able to select the value from a drop-down list.	**Design-level**	Chooses a particular widget.
10	The product screens shall look as shown in appendix xx.	**Design-level**	Prescribes the screen design.

Order of options may differ.

2.2: usability requirements and contractual risk (10 points; PDF pp. 26-27)

Question: In a bespoke project building a tailor-made solution, decide whether each requirement mainly poses a risk for the customer or supplier, then select the most valid offered reason.

Item	Requirement as in PDF	Suggested party	Select this offered reason	Explanation
R1	At most 1 of 5 novices shall encounter critical problems during tasks Q and R. At most 5 medium problems shall be encountered in total.	**Supplier**	**It cannot be known whether this requirement is feasible.**	Before trying a prototype with representative users, the supplier cannot safely promise those problem-count limits. The limits depend on the chosen tasks and population.
R2	Novice users shall perform tasks Q and R in 15 minutes. Experienced users shall perform tasks Q, R, and S in 2 minutes.	**Supplier**	**The requirement cannot be assessed early in the project.**	Reliable assessment requires a functional prototype and experienced users; mockups and think-aloud tests distort task-time measurements. The supplier therefore risks committing to limits whose feasibility cannot be assessed early.
R3	Recording breakfast shall be possible with 5 keystrokes per guest. No mouse.	**Customer**	**The solution might not work for the customer even if the requirement is fulfilled**	Few keystrokes do not guarantee learnability, understandable controls or successful real work.
R4	System shall use screen pictures in app. xx, buttons work as app. yy	**Customer**	**The solution might not work for the customer even if the requirement is fulfilled**	A supplier can implement the prescribed interface exactly while it still fails to support the customer's tasks.
R5	The system shall follow style guide zz. Menus shall have at most three levels.	**Customer**	**The solution might not work for the customer even if the requirement is fulfilled**	Conformance to a style guide and a menu-depth limit does not establish usable task support.

Part 3: essay questions and model answers

Shared question instructions: Write an essay that includes all listed concepts. Assessment includes both the overall description/argument and whether the concepts are explained or exemplified in sufficient depth. Answers should be clear, readable and in English.

3.1: Elicitation (10 points; PDF p. 28)

Question: Write an essay on **Elicitation**, covering these concepts: 3 examples of elicitation techniques (including how they are used, when to use them, when not to use them, if and how they should be combined); stakeholder; negotiation of conflicting requirements.

Length: 750 words / one page.

Model answer Requirements elicitation discovers the needs, goals and constraints of stakeholders: people, groups or organisations affected by or interested in the proposed system. A hotel project should consider reception staff, managers, guests, maintenance staff and connected service providers. Stakeholder identification must include people with little decision power, since they can still experience substantial consequences.

Interviews are useful early, when the analyst needs to understand goals, vocabulary and existing work. Ask open questions, request concrete examples and clarify unfamiliar terms immediately. Asking why exposes the goal behind a requested feature. Interviews are costly for large populations, rely on what people can articulate, and are a weak standalone way to resolve conflicts between groups. Avoid leading questions and premature solution proposals.

Questionnaires become useful once initial interviews have established enough domain understanding to write meaningful questions. They can compare priorities across many hotel employees or investigate how often a discovered problem occurs. Pilot the questionnaire and sample the relevant groups. Avoid relying on questionnaires for poorly understood exploratory issues: respondents cannot normally ask clarifying questions, and badly chosen wording or response options can hide important needs.

Prototypes with representative task trials help later, once candidate workflows exist. Ask receptionists to book, amend and cancel reservations while observing their actions. A simple prototype can reveal misunderstood requirements before implementation is expensive. Static screen approval alone is insufficient. A visual prototype is also weak evidence for production capacity or reliability unless those properties are deliberately represented and measured.

The techniques complement each other: interviews establish problems and vocabulary, questionnaires investigate their prevalence and priority, and prototype trials validate proposed support. Trial findings can trigger further interviews, so these phases are iterative rather than a rigid sequence.

Conflicting requirements require negotiation. For example, a manager may request extensive booking checks while receptionists need rapid check-in. Identify the underlying goals, affected stakeholders and constraints; agree measurable criteria and priorities; compare alternatives such as performing some checks earlier. A facilitated joint discussion can negotiate an acceptable trade-off. Record the decision, rationale and links to goals, then validate it with the affected groups. A powerful stakeholder's preference alone does not establish that the compromise meets the project's needs.

Grading The model answer above is newly written with help of AI, using the prompt, planning hints and course material; it is not a copied student answer or an official marking scheme.

Here is the markign scheme/allocation of 10 points:

- topic (2)
- five concept components (5)
- demonstrated understanding (1)
- flow/combination (2)

3.2: Checking and Validation (10 points; PDF p. 29)

Question: Write an essay on **Checking and Validation**, covering these concepts: complete; unambiguous; consistent; modifiable; verifiable.

Length: 750 words / one page.

Model answer Checking and validation reduce different risks. Requirements checking examines whether the specification is well formed and internally sound. Validation asks whether it expresses the right needs and supports stakeholders' goals. A perfectly consistent specification can still describe the wrong product, so both activities should start early and recur as requirements change.

A **complete** specification covers the relevant needs, tasks, variants, data and qualities. A content checklist can find missing sections, and a CRUD matrix can reveal data without supporting operations. However, filling a template does not prove completeness. Walk through realistic and exceptional tasks with representative stakeholders, such as late hotel arrival or cancellation after payment, to discover omitted needs.

An **unambiguous** requirement has one intended interpretation. Inspect wording, define terms in a glossary and ask different readers to paraphrase the requirement. "Handle bookings quickly" is ambiguous; an agreed response-time measure under a defined workload is clearer. Validate the chosen interpretation with users, rather than merely checking that a sentence looks precise.

A **consistent** specification contains no incompatible obligations and agrees across its views. Compare requirements, task descriptions and data models. If one requirement says cancelled bookings must be retained and another requires their immediate removal, resolve the apparent contradiction, possibly by distinguishing active records from history. CRUD checks help uncover mismatches between data entities and specified tasks.

A **modifiable** specification supports changes without introducing avoidable contradictions. Use stable identifiers, a clear structure, controlled versions and limited duplication. Trace a booking-retention rule to related functions and tests, then use those links to assess a proposed change. Inspect whether the change has been made everywhere it matters and validate its consequences with affected stakeholders.

A **verifiable** requirement has a feasible way to determine whether it is met. Specify conditions, measures and acceptance thresholds. For example, measure the 95th-percentile booking-search response time under an agreed workload, with an agreed maximum. Review the planned test before committing to the requirement. Passing that test checks compliance; establishing that its workload and threshold address actual reception needs is validation.

Inspections, stakeholder reviews, walkthroughs and prototype trials should therefore be combined. Their purpose is to reveal defects and wrong assumptions while correction is still inexpensive, rather than to claim that any single checklist proves the specification correct.

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Here is the markign scheme/allocation of 10 points:

- describe the topic (1)
- explain the five concepts (5)
- exemplify checking versus validation (1)
- explain checks/validation for more than half the concepts (3)